Object-Oriented and Classical Software Engineering

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CHAPTER 3

THE SOFTWARE PROCESS

Introduction

- The software process is *the way we produce software*
- It incorporates
  - The methodology with its underlying software life-cycle model
  - Techniques and tools we use
  - The individuals building the software
- Different organisations have different software processes

3.1 The Unified Process

- Methodology is one component of a software process

- The primary object-oriented methodology used today is the ***Unified Process***

The Unified Process (contd)

- The Unified Process is *not* a specific series of steps for constructing a software product
  - No such single "one size fits all" methodology could exist, because there is a wide variety of different types of software
- The Unified Process is an adaptable methodology
  - It has to be modified for the specific software product to be developed

3.3 The Requirements Workflow

- Development process starts when a client approaches a development organisation

- The aim of the requirements workflow
  - To determine the client’s needs

Overview of the Requirements Workflow

- First, gain an understanding of the *application domain* (or *domain*, for short)
  - That is, the specific business environment in which the software product is to operate
- Second, build a business model
  - A document that demonstrates the cost effectiveness of the software
  - If at any time the client does not feel that the cost is justified, development terminates immediately

Overview of the Requirements Workflow

- It is vital to determine what the client needs and to find out what constraints exist
  - Deadline
    - Nowadays, software products are often mission critical
  - Reliability
  - Cost
    - The client will rarely inform the developer how much money is available
    - A bidding procedure is used instead

Overview of the Requirements Workflow

- Preliminary investigation of the client's needs is often called *concept exploration*

- The aim of this *concept exploration* is to determine
  - What the client needs
  - *Not* what the client wants

3.4 The Analysis Workflow

- The aim of the analysis workflow
  - To analyze and refine the requirements
- Why not do this during the requirements workflow?
  - The requirements artifacts must be totally comprehensible by the client
- The artifacts of the requirements workflow must therefore be expressed in a natural (human) language
  - All natural languages are vague

The Analysis Workflow

- Ambiguities are unlikely to arise if a mathematical language is used
  - Client is unlikely to understand
- Two separate workflows are needed
  - The requirements artifacts must be expressed in the language of the client
  - The analysis artifacts must be precise, and complete enough for the designers
  - More detail is added in the analysis workflow which are not relevant to the client’s understanding of product

The Specification Document

- Specification document (“specifications”)
  – It constitutes a contract
  – It must not have vague phrases like “optimal,” or “98% complete”, “suitable”, “convenient”, “ample” or “enough”

- Having complete and correct specifications is essential for
  – Testing and
  – Maintenance

The Specification Document

- Unless specifications are precise it cannot be determined
  - Whether they are correct
  - Whether the product meets the specifications
- With the unified process, a specification document consists of UML diagrams

The Specification Document

- During analysis, the architecture of the product is determined
  - Classes
  - Attributes (data) of classes

Software Project Management Plan

- Once the client has signed off the specifications, detailed planning and estimating begins
- We draw up the software project management plan (SPMP), including
  - Cost estimate
  - Duration estimate
  - Deliverables
  - Milestones
  - Budget
- This is the earliest possible time for the SPMP

3.5 The Design Workflow

- Specifications
  - What the product must do
- Design
  - How the product is to do it
- The aim of the design workflow is to refine the artifacts of the analysis workflow until the material is in a form that can be implemented by the programmers

Object-Oriented Design

- Basis of the paradigm is the *class*
  - Classes are extracted during the analysis workflow, but
  - Designed during the design workflow
  - Methods are extracted and assigned to classes
  - Interfaces between methods are determined
  - Algorithms are chosen for each method
  - Attributes are extracted during analysis but formats are assigned during design

3.6 The Implementation Workflow

- The aim of the implementation workflow is to implement the target software product in the selected implementation language
  - A large software product is partitioned into subsystems (implemented in parallel by coding teams)
  - The subsystems consist of *components* or *code artifacts* (implemented by an individual programmer)

- Usually the programmer is only given the design artifact

3.7 The Test Workflow

- In the Unified Process, testing is carried out in parallel with the other workflows.

- The test workflow is the responsibility of
  - *Every* developer and maintainer, and
  - The quality assurance group

- Traceability of artifacts is an important requirement for successful testing

3.8 Postdelivery Maintenance

- Postdelivery maintenance is an essential component of software development
  - More money is spent on postdelivery maintenance than on all other activities combined

- Problems can be caused by
  - Lack of documentation of all kinds

Postdelivery Maintenance

- Two types of testing are needed
  - Testing that the required changes made during postdelivery maintenance have been implemented correctly
  - Regression testing by using previous test cases (while making required changes, no other inadvertent changes were made)

- All previous test cases (and their expected outcomes) need to be retained

3.9 Retirement

- Software can be unmaintainable because
  - A drastic change in design has occurred
  - So many changes have been made to the original design that interdependencies occur and even a small change to one component might have a drastic effect on the functionality of the product as a whole.
  - Documentation is missing or inaccurate
  - Hardware is to be changed — it may be cheaper to rewrite the software from scratch than to modify it

Retirement

- True retirement is a rare event
- It occurs when the client organization no longer needs the functionality provided by the product

3.10 The Phases of the Unified Process

- The increments are identified as phases

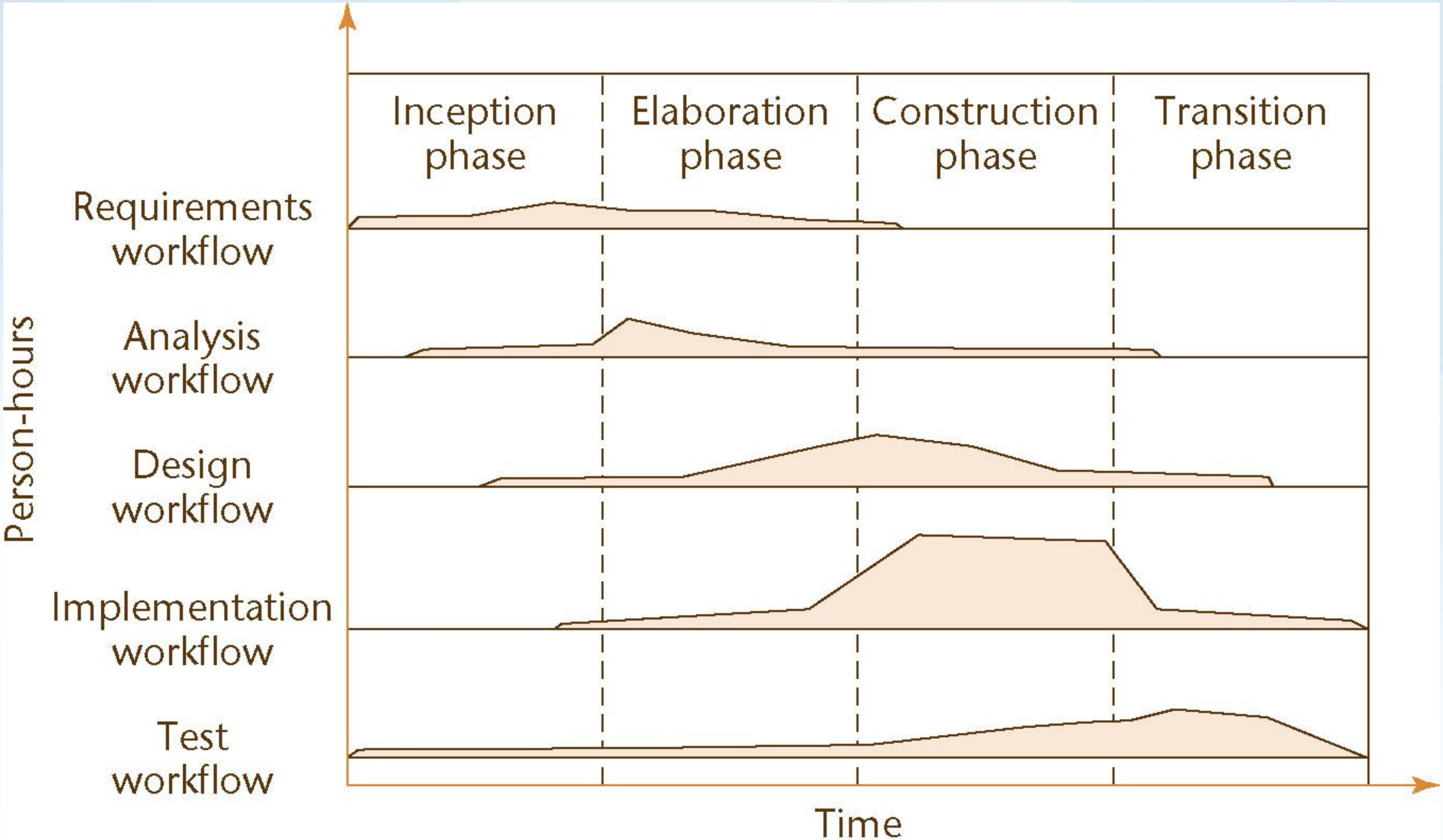

Figure 3.1

The Phases of the Unified Process

- The four increments are labeled
  - Inception phase
  - Elaboration phase
  - Construction phase
  - Transition phase

- The phases of the Unified Process are the increments

The Phases of the Unified Process

- In theory, there could be any number of increments
  - In practice, development seems to consist of four increments
- Every step performed in the Unified Process falls into
  - One of the five core workflows and *also*
  - One of the four phases

3.10.1 The Inception Phase

- The aim of the inception phase is to determine whether the proposed software product is economically viable

The Inception Phase

1. Gain an understanding of the domain
2. Build the business model
   – Understand how the client organisation operates in that domain
3. Delimit the scope of the proposed project
   – Focus on the subset of the business model that is covered by the proposed software product
4. Begin to make the initial business case

The Inception Phase: Analysis, Design Workflows

- A small amount of the analysis workflow may be performed during the inception phase
  - Information needed for the design of the architecture is extracted

- Accordingly, a small amount of the design workflow may be performed, too

The Inception Phase: Implementation Workflow

- Coding is generally not performed during the inception phase

- However, a *proof-of-concept prototype* is sometimes build to test the feasibility of constructing part of the software product

The Inception Phase: Test Workflow

- The test workflow commences almost at the start of the inception phase
  - The aim is to ensure that the requirements have been accurately determined

The Inception Phase: Planning

- There is insufficient information at the beginning of the inception phase to plan the entire development
  - The only planning that is done at the start of the project is the planning for the inception phase itself
- For the same reason, the only planning that can be done at the end of the inception phase is the plan for just the next phase, the elaboration phase

3.10.2 Elaboration Phase

- The aim of the elaboration phase is to:
  - Refine the initial requirements
  - Refine the architecture
  - Monitor the risks and refine their priorities
  - Refine the business case
  - Produce the project management plan

- The major activities of the elaboration phase are refinements or elaborations of the previous phase

The Tasks of the Elaboration Phase

- The tasks of the elaboration phase correspond to:
  - All but completing the requirements workflow
    - Chapter 11
  - Performing virtually the entire analysis workflow
    - Chapter 13
  - Starting the design of the architecture
    - Section 8.5.4

3.10.3 Construction Phase

- The aim of the construction phase is to produce the first operational-quality version of the software product
  - This is sometimes called the beta release

The Tasks of the Construction Phase

- The emphasis in this phase is on
  - Implementation and
  - Testing
    - Unit testing of modules
    - Integration testing of subsystems
    - Product testing of the overall system

3.10.4 The Transition Phase

- The aim of the transition phase is to ensure that the client’s requirements have indeed been met
  - Faults in the software product are corrected
  - All the manuals are completed
  - Attempts are made to discover any previously unidentified risks
- This phase is driven by feedback from the site(s) at which the beta release has been installed

Capability Maturity Models: SW–CMM

- A strategy for improving the software process
- Fundamental ideas:
  - Improving the software process leads to
    - Improved software quality
    - Delivery on time, within budget
  - Improved management leads to
    - Improved techniques

Capability Maturity Models: SW–CMM

- Five levels of *maturity* are defined
  - Maturity is a measure of the goodness of the process itself
- An organization advances stepwise from level to level

Level 1. Initial Level

- No sound software engineering management practices are in place in the organisation
- Ad hoc approach
  - The entire process is unpredictable
  - Management consists of responses to crises
- Most organizations world-wide are at level 1

Level 2. Repeatable Level

- Basic software management practices are in place
  - Management decisions should be made on the basis of previous experience with similar products
  - Measurements (“metrics”) are made
  - These can be used for making cost and duration predictions in the next project
  - Problems are identified, immediate corrective action is taken

Level 3. Defined Level

- The software process is fully documented
  - Managerial and technical aspects are clearly defined
  - Continual efforts are made to improve quality and productivity
  - Reviews are performed to improve software quality

Level 4. Managed Level

- Quality and productivity goals are set for each project
  - Quality and productivity are continually monitored
  - Statistical quality controls are in place to enable management to distinguish a random deviation from a meaningful violation of quality or productivity standards.

Level 5. Optimizing Level

- Continuous process improvement
  - Statistical quality and process controls techniques are used to guide the organisation
  - Feedback of knowledge from each project to the next

Experiences with SW–CMM

- It takes:
  - 3 to 5 years to get from level 1 to level 2
  - 1.5 to 3 years from level 2 to level 3